Product assortment (example)

	Display	Battery	Price	Utility
A Zentria CoreBook*	2.8K OLED	12h	$480	**0.6411**
B Acer Aspire 5 A515	FHD IPS	8h	$580	0.4989
C ASUS VivoBook 15	FHD OLED	7h	$680	0.4566
D Dell Inspiron 15	FHD	6h	$700	0.3741
E HP Pavilion 15	FHD	7h	$730	0.3705

Utility is computed by the researcher and never shown to the model.

*Fictional brand with the highest utility score. 34 assortments, 20 categories

Request conditions

(32 conditions)

Requests: Open request · Vague request · Priorities in order

Also: Exact requirements · Brand ruled out

Brands: Names shown · Names hidden

Checks: Descriptions removed · Price changed · Scores supplied

Large language models

(30 models, 7 developers)

Anthropic: Haiku 4.5 · Sonnet 4.6 · Opus 4.6 · Opus 4.7

OpenAI: GPT-4o · 4o Mini · 4.1 Mini · 4.1 Nano · 5 Mini · GPT-5.4 · 5.4 Mini · 5.4 Nano

Google: 2.0 Flash · 2.5 Flash · 2.5 FL · 2.5 Pro · 3 Flash · 3.1 Pro · 3.1 FL · Gemma 3 27B · Gemma 4 31B

Open-wt: LLaMA 70B · DS V3 · DS R1 · Qwen 72B · Kimi K2

Coding of explanations

generates → Response → evaluates (same model)

- **Coherence** 0-100
- **Acknowledgment of criteria** 0-100
- **Brand reasoning** yes / no

Classification:

Feature-only explanation = worse product, no brand reason

"I'm looking for a reliable laptop for everyday work and browsing ..."

The rest of the prompt varies by condition:

Open request

"Which product do you recommend?"

25.0%

non-optimal, names shown
15.1% with names hidden

Stated priorities

"I want something reliable ..." and "1. Battery life and portability ..."

22.4% / 17.4%

vague request,
priorities in order

Promotional training

Fine-tuning on promotional examples for one brand, primary of eight runs

52.5%

recommend the trained brand, from 32.0%

Activation steering

The internal signal changed at nine strengths

57.4 to 78.4%

non-optimal recommendations

Exact requirements

People and models given the same product requirements

99.2%

of model decisions,
against 69.1% of people

Consumer studies

Stated product choices after a recommendation, in static and live shopping

N = 3,164

participants across four studies

"The ASUS VivoBook 15 is the best option as it offers *a blend of high performance and reliability. ... Its vibrant FHD OLED display and solid 7-hour battery life make it ideal for everyday usage ..."* GPT-4o

Feature-only explanation:

Product A has superior display (2.8K OLED), longer battery (12h), and lower price ($480). Cited attributes are matched or beaten by product A.

74% of worse recommendations give no brand reason

Brand names change recommendations

Non-optimal recommendations in all 30 models;

brand names raise the rate in 29 of 30